PART 2

MODUPERF TAPE PERFORATOR MANUAL

901-1951
PART II

		Page
1.	SCOPE	1-1
2.	GENERAL NOTES	1-1
2.1	Unpacking Instructions	1-1
2.2	Packing Instructions	1-1
2.3	Recommended Tape Material	2-1
2.4	Definitions of References	2-2
2.5	Location of Serial Numbers	2-2
3.	OPERATIONAL DESCRIPTION	3-1
3.1	Outline of Operation	3-1
3.2	Detailed Operation	3-1
3.2.1	Punch Pin Selection	3-1
3.2.2	Tape Punching	3-1
3.2.3	Tape Advance	3-3
4.	PREVENTIVE MAINTENANCE	4-1
4.1	Preventive Maintenance Interval	4-1
4.2	Lubricants	4-1
4.3	Tool List	4-1
4.4	Preventive Maintenance Procedure	4-1
4.4.1	Procedure Outline	4-1
4.4.2	Removal/Remounting of Selector Module	4-2
4.4.3	Prev. Maintenance of Selector Module	4-3
4.4.4	Prev. Maintenance of Die Block Module	4-3
4.4.5	Prev. Maintenance of Main Frame Module	4-4
5.	CORRECTIVE MAINTENANCE	5-1
5.1	General	5-1
5.2	Sequence of Adjustments	5-1
5.3	Tool List	5-1
5.4	Detailed Adjustments	5-2

5.4.1 Mag. Pick-up Gap 5-2
5.4.2 Timing Disc Location 5-3
5.4.3 Sprocket Drive Assembly 5-5
5.4.4 Feed Sprocket Position 5-6
5.4.5 Tape Lid Gap 5-7
5.4.6 Tape Guide 5-8
5.4.7 Belt Tension - Stepper Motor 5-9
5.4.8 Belt Tension - Drive Motor 5-10
5.4.9 Character Spacing 5-11

5.5 Tape Supply Requirement 5-12

6. FAULT ANALYSIS 6-1

6.1 General 6-1
6.2 Fault Analysis Chart 6-1

7. FIELD MODIFICATIONS 7-1

7.1 General 7-1
7.2 Tape Guide Setting Change 7-1
7.3 Die Block Change 7-1
7.4 Selector Module Change 7-3

1. SCOPE

This document is intended to assist the technician in maintaining the MODUPERF Tape Perforator Mechanism. Instructions include preventive and corrective maintenance procedures. These should be thoroughly understood before attempting to service the equipment.

2. GENERAL NOTES

2.1 UNPACKING INSTRUCTIONS

When removing the Perforator Mechanism or other maintenance parts/modules from the carton, be careful not to lift or carry the mechanism by fragile parts which could be damaged. The units have been carefully tested, inspected, and packed to insure arrival in good condition.

Inspect the unit immediately after removal from the packing container to determine if any damage has occured in transit. In the event the equipment has been damaged as a result of shipping, the carrier and Data I/O must be notified immediately.

2.2 PACKING INSTRUCTIONS

If the instrument is to be shipped to Data I/O for service or repair, attach a tag to it describing the work required and identifying the owner. In correspondence, identify the unit by serial number, model number and name.

If the original shipping container is to be used, place the instrument in the container with appropriate packing material and seal the container well, with strong tape. If some other container is used, be sure that it is a heavy carton, wrapped with heavy paper or plastic; use appropriate packing material and seal well with strong tape. Mark the container "DELICATE INSTRUMENT" or "FRAGILE."

DO NOT package any equipment in a rigid container without utilizing shockmounts or shock absorbant packing material. A rigid container will allow shock on the outside to be transmitted undamped to the mechanism which may cause damage.

Before a complete perforator mechanism or a die block module is packaged, the chad chamber above the die block should be cleared of chad. This will prevent chad from spreading thruout the unit during the shipment.

2.3 RECOMMENDED TAPE MATERIALS

PAPER TAPE: Oiled* or unoiled colored or buff tape is recommended. Black tape should be avoided since sometimes carbon dye is used which is very detrimental to the wear characteristics of the punch pins. The tape should be from a quality manufacturer and meet EIA Standard RS-227-A.

*CAUTION: Oiled buff tape is not recommended if the tape will be read with a photo-optical reader. The 5 to 12% oil in these tapes sometimes causes highly transparent spots to occur. These spots, combined with the light yellow "buff" color, may exceed 55% light transmissivity and cause reader errors. While oiled tapes of gray and darker colors are acceptable since their transmissivity is well below 55%, the oil in the tape is of no value since it has an insignificant effect on the life expectancy of the MODUPERF punch head.

LIFE EXPECTANCY OF PUNCH HEAD:

Paper Tape; any color but black	500,000 feet of tape

PAPER tapes, if of poor quality, will cause problems in punching and/or reading. For example, the tape might have the following defects; transparent spots, width out of tolerance, spliced in roll not continuous, torn folds (fanfold) etc. The tapes of the suppliers listed below have been tested and found to be acceptable. This list is not meant to be exclusive.

Arvey Corporation
3500 N. Kimball Ave.
Chicago, IL 60618
(312) 463-1400

Chase Foster
199 Amaral St.
East Providence, RI 02914
(401) 434-2340

Eastern Specialties
P.O. Box 350
Holyoke, MA 01040
(413) 533-7103

Numeridex, Inc.
241 Holbrook Drive
Wheeling, IL 60090
(312) 541-8840

Occidental
1188 Montague View
San Leandro, CA 94577
(415) 352-3202

Paper Manufacturers Company
9800 Bustleton Avenue
Philadelphia, PA 19115
(215) 673-4500

Ribco
200 Railroad Avenue
Paterson, NJ 07509
(201) 271-4848

2.4 DEFINITIONS OF REFERENCES

References made to "Left", "Right", "Up", "Down", "Front", "Rear", etc. apply to the unit in its normal operating position as viewed from the front, ie; the tape moves from left to right, and code level 1 on the tape is nearest to the front.

2.5 LOCATION OF SERIAL NUMBERS

The Data I/O 950-1951 has a serial number located on the label on its back panel. Use this number in correspondence with Data I/O regarding the equipment.

The perforator mechanism consists of 3 main modules, the main frame module, the selector module, and the die block module. Each module has a unique serial number. The serial number of the main frame module is also used to define the complete perforator mechanism. For example, a perforator mechanism with a main frame module S/N 1234 is defined as perforator mechanism S/N 1234. The serial numbers are located as follows:

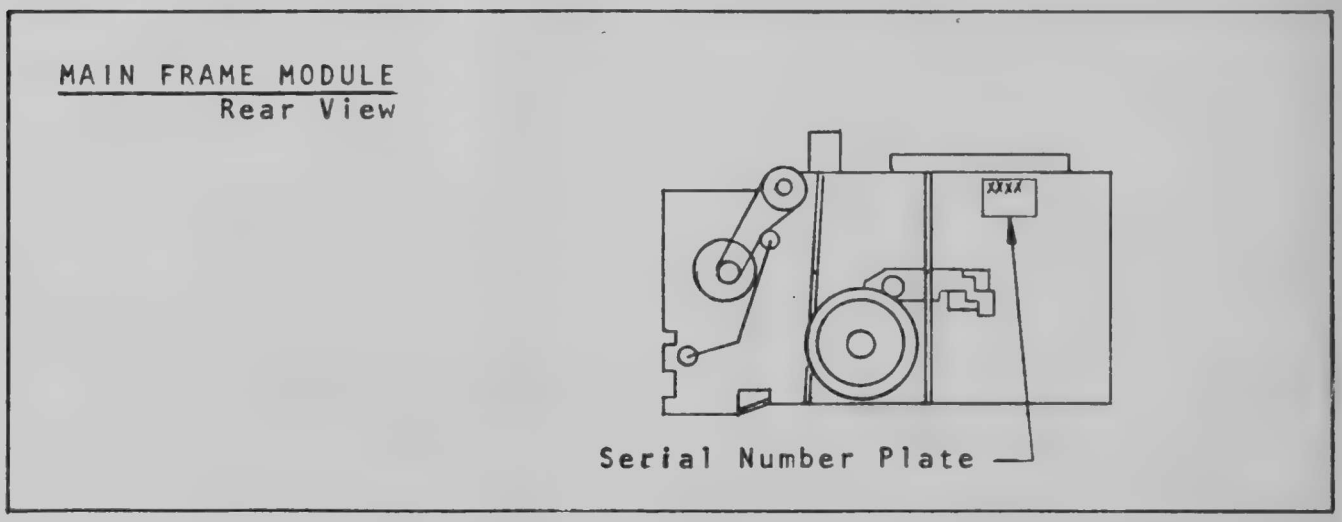


Figure 2-1

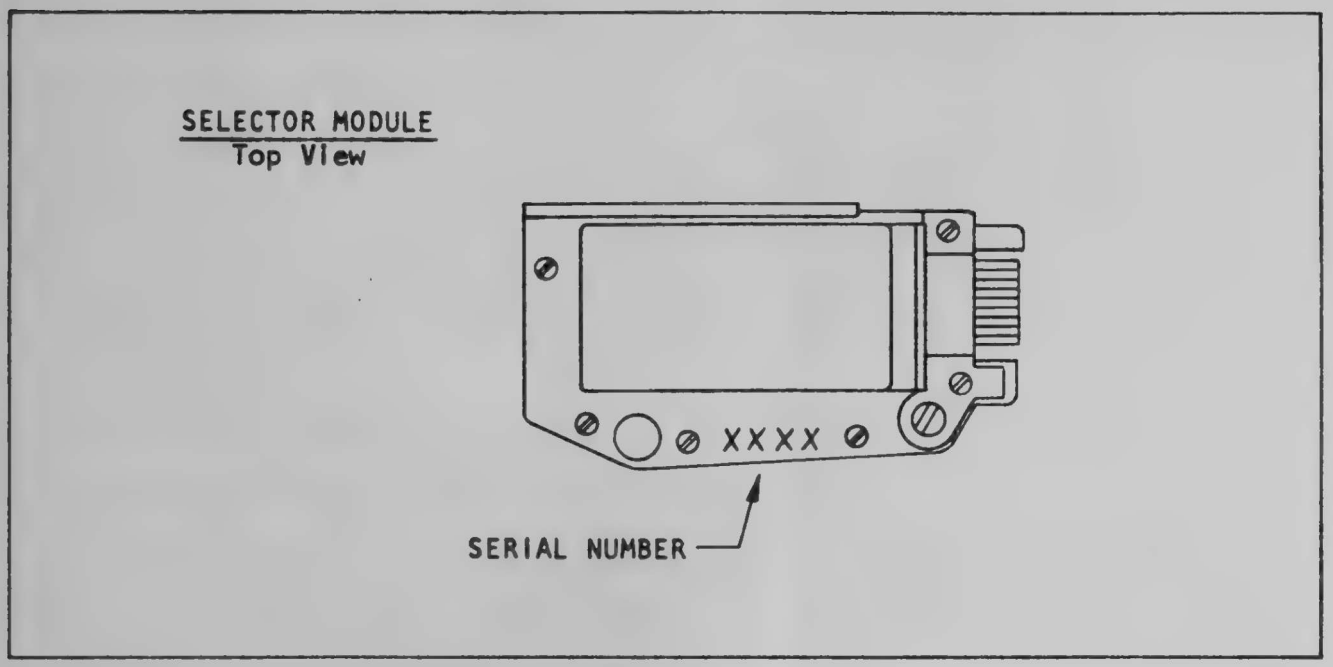


Figure 2-2

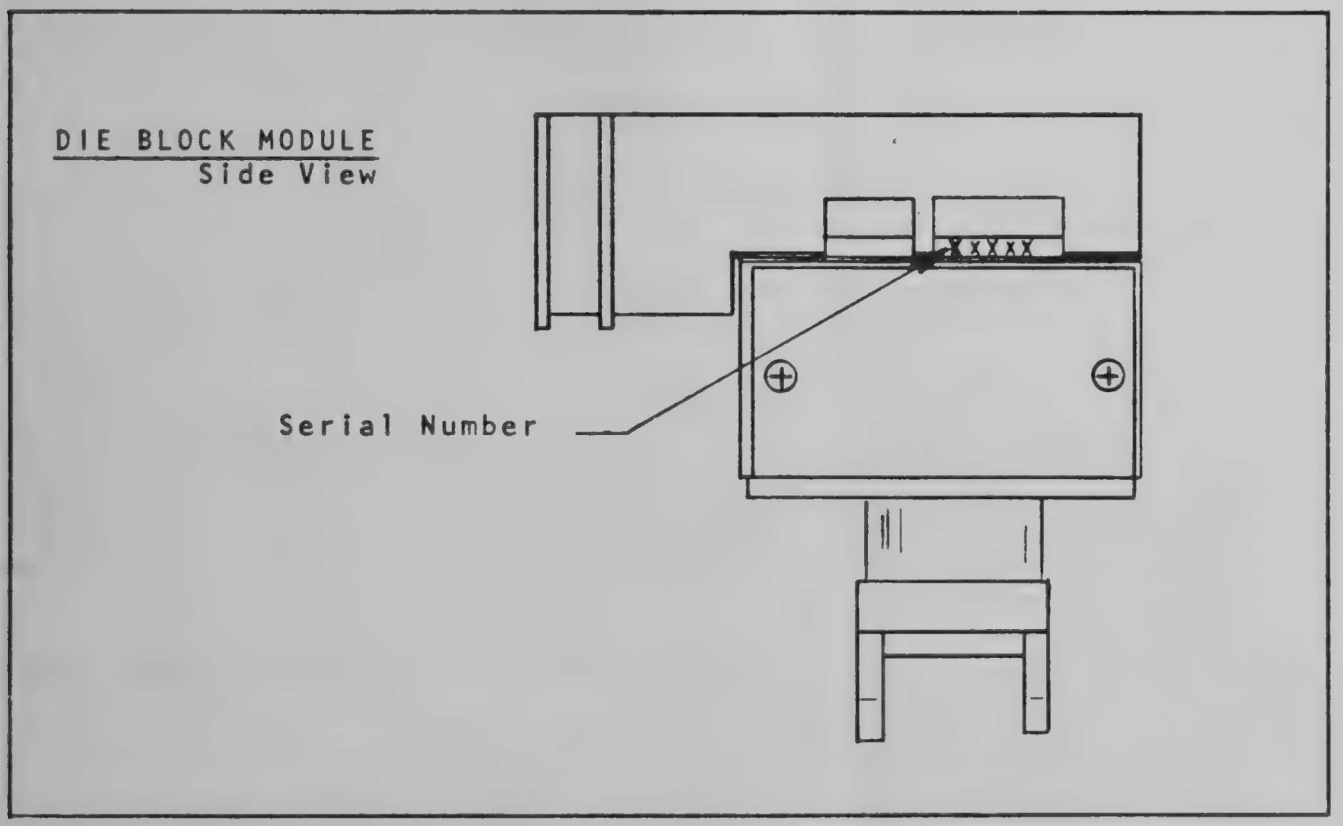


Figure 2-3

3. OPERATIONAL DESCRIPTION

3.1 OUTLINE OF OPERATION

The **MODUPERF** Tape Perforator is a device which perforates holes in paper tape in response to electrical drive signals. Electrical pulses received by the selector module are converted into a mechanical arrangement of the selecting slides. These slides engage with the punch pins and are driven through the paper by a motor. After punching the tape is advanced by a stepper motor.

3.2 DETAILED OPERATION

3.2.1 PUNCH PIN SELECTION (see Figure 3-1)

The code channels and feed track punches are selected by the selector module. The selecting module consists of electromagnets and selecting slides. There is one electromagnet (coil) and one selecting slide for each punch pin.

When a coil is activated by an electrical signal, an armature lever is drawn to the core of the coil. Attached to the armature lever is a wire which engages with a selecting slide. As the armature lever is drawn in to the core, the wire moves the selecting slide to the right where it engages with the punch bar.

Once the hole has been punched (see Section 3.2.2) and the next tape position does not call for a hole, the selecting coil is deactivated and the selecting slide disengages the punch bar by moving to the left. If consecutive holes are being punched, the selecting slide stays engaged with the punch bar.

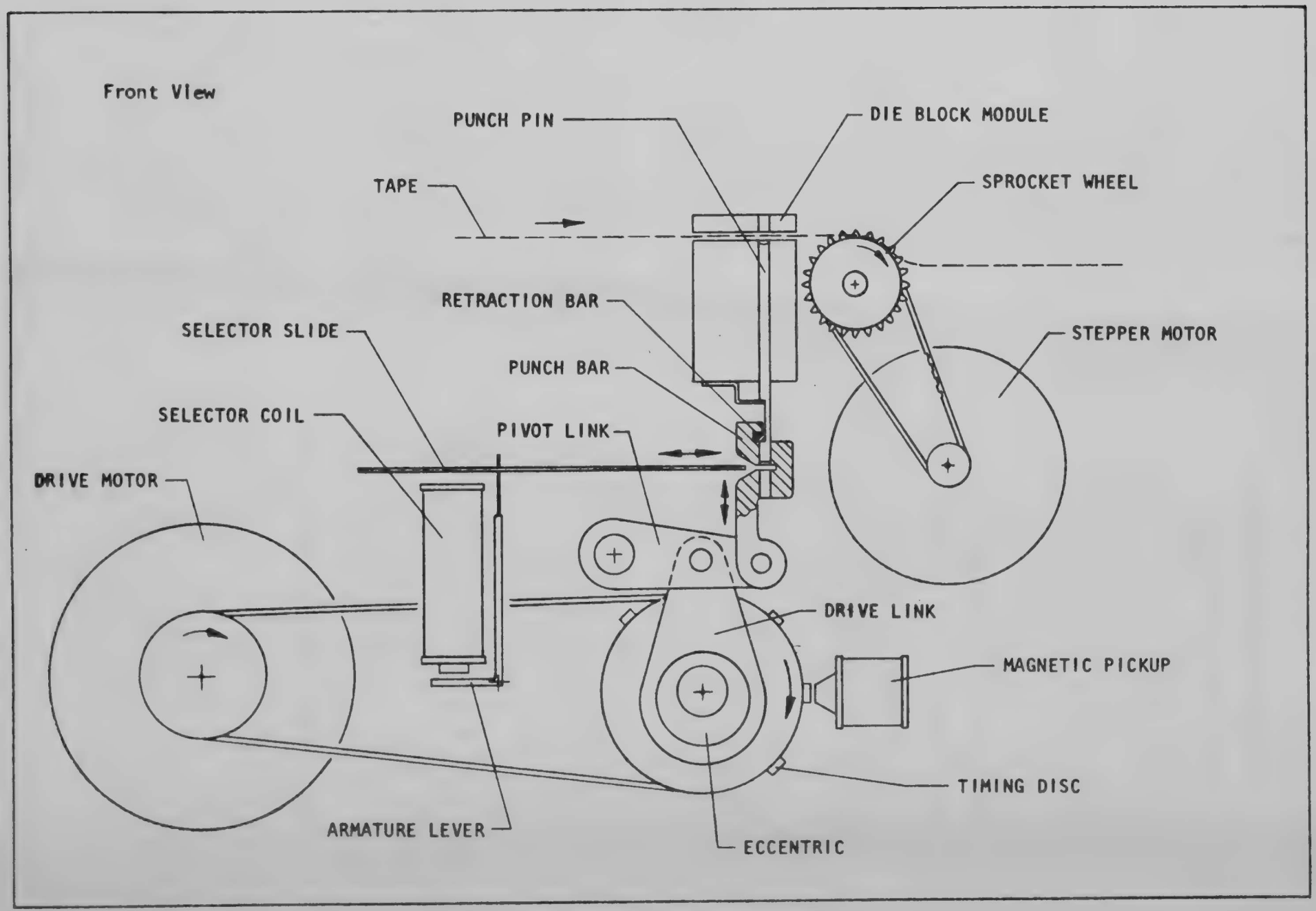
This "Non Return to Zero" operating principle drastically reduces the motion of the selecting slides. For example, if 10 consecutive holes are punched, the slide moves to the right only once, before the first hole, and returns to the left only once, after the tenth hole. It naturally follows that the selecting slide for the feed track gets activated only once for any string of data.

3.2.2 TAPE PUNCHING (see Figure 3-1)

An AC drive motor is used to supply the energy to push the punch pins through the paper and then retract the pins. The rotary motion of the motor is converted to simple harmonic motion by belting the motor to an eccentric crank. Since the punch bar is connected to the crank, it also moves up and down in simple harmonic motion. The motion of the selecting slide is synchronized with the motion of the punch bar by employing a magnetic pick-up which senses the position of the mainshaft with the eccentric crank.

The timing is set so the selecting slide enters the punch bar when the eccentric is at its low point. The punch bar then

FIGURE 3-1 TAPE PERFORATOR MECHANISM



moves upward. This action causes the selecting slide to make contact with the bottom surface of the punch pin. Further motion of the punch bar causes the punch pin to be forced through the paper that is threaded thru the die block assembly. It should be noted that the selecting slide is flexing while this action is taking place.

When the eccentric is at top dead center, the punch pins have completed the upward motion. Further motion of the eccentric causes the retraction bar to engage with the lower corner of the notched section in the punch pin, resulting in positive retraction from the tape. During the down stroke of the punch bar, the selecting slides are removed from their selected position.

The sequences described above cause holes to be punched in the tape. If the selecting slide is not activated, the punch bar motion will not push the selecting slide into the punch pin. Thus, the punch pin will not move, and tape will not be punched.

3.2.3 TAPE ADVANCE (see Figure 3-1)

The feed track in the tape is engaged with a sprocket wheel. The sprocket wheel is belted to a stepping motor. The motor is signaled by the magnetic pick-up to advance the tape when the punch pins leave engagement with the tape.

To backspace, the stepper motor is directed to step in the reverse direction.

4. PREVENTIVE MAINTENANCE

4.1 PREV. MAINTENANCE INTERVAL

Preventive maintenance consists primarily of cleaning and lubricating the perforator mechanism. It should be performed as a regular routine to insure maximum life and reliability.

Paper Tape, any color but black:	12 months, or 250 1000 ft. rolls, whichever comes first.

The environmental operating conditions, as well as the tape material, must be considered before applying the above periods. See Section 2.3 for information on recommended tape materials.

4.2 LUBRICANTS

The widely used Western Electric KS7470 oil and KS7471 grease are suitable for the MODUPERF mechanism. They are available from the following sources:

Part No./Description		Supplier
KS7470	Oil	Western Electric
KS7471	Grease	
88970	Oil (1 qt. can)	Teletype Corp.
88971	Oil (1 gal. can)	Skokie, Illinois
88973	Grease (1 lb. can)	
7668	Oil, Lub. Kit, (4 oz. bottle)	Data Specialties, Inc.
7669	Grease, Lub. Kit. (2 oz. jar)	Northbrook, Illinois

4.3 TOOL LIST

The following tools are required to perform preventive maintenance on the perforator mechanism.

5/16" Nut Driver, outside diameter at socket to be .450" maximum.

Small soft bristle brush, to remove paper lint from mechanism.

4.4 PREVENTIVE MAINTENANCE PROCEDURE

4.4.1 PROCEDURE OUTLINE

The MODUPERF perforator mechanism consists of 3 main modules; main frame module, die block module, and selector module. To service the unit, the selector module is removed to gain access to some of the lubricating points.

Due to the unique construction of the mechanism, the removal of the selector module is very easy, fast, and requires only the removal of 2 screws. Similarly, the remounting is also easily accomplished and NO ADJUSTMENTS are necessary. The objective of the preventive maintenance procedure is to clean, lubricate, and visually inspect the mechanism.

4.4.2 REMOVAL/REMOUNTING OF SELECTOR MODULE

Removal:

1. Loosen 2 screws (5/16" Hex Head) which hold selector module to main frame, and slide selector module to its maximum left position.

2. Remove 2 screws and selector module from unit.

Remount:

1. Mount selector module to main frame in its maximum left position. Do not tighten the 2 screws.

2. Slowly slide selector module to the right until it gets stopped by pin protruding from die block module.

3. While pressing selector module against pin, tighten 2 screws. Make sure once screws are tightened, the selector module firmly rests against pin.

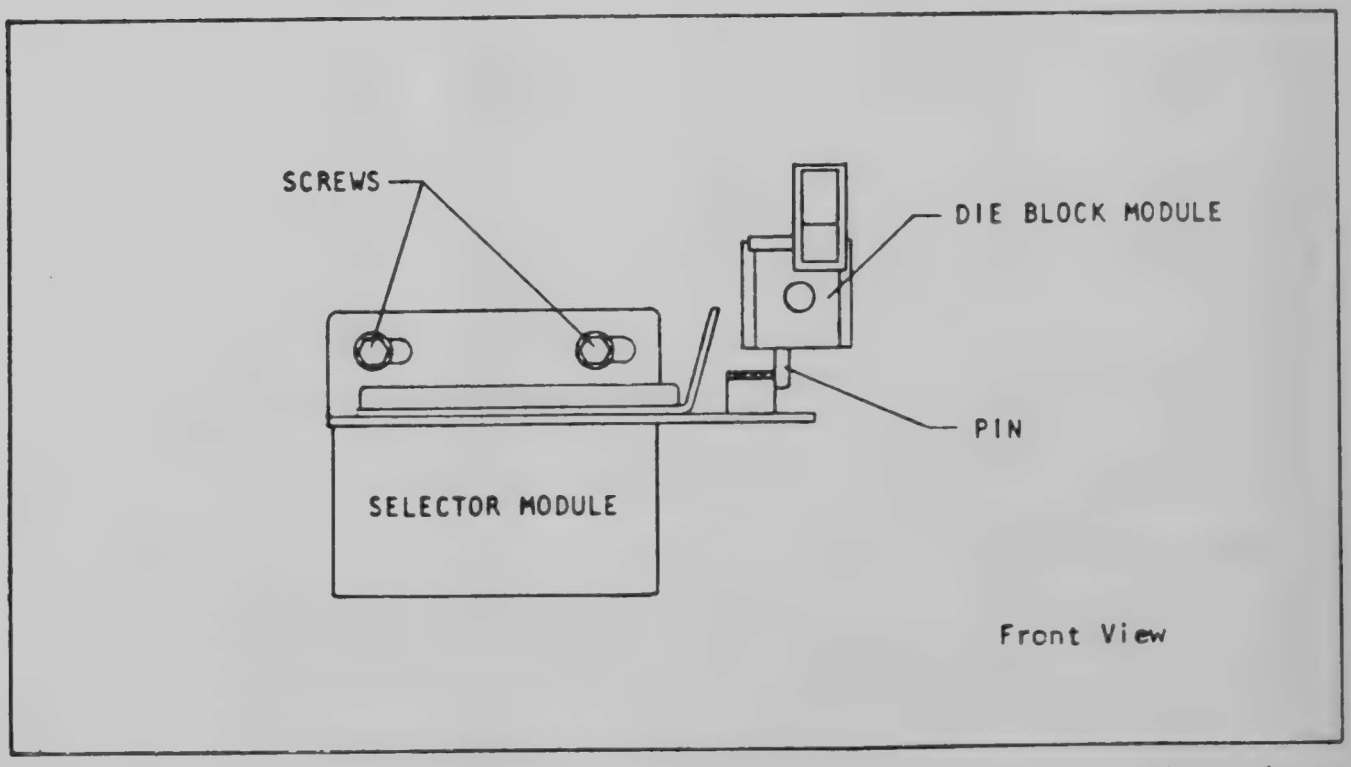


Figure 4-1

4.4.3 PREVENTIVE MAINTENANCE OF SELECTOR MODULE

Remove paper lint and excess grease. Inspect module for damaged or loose parts. DO NOT LUBRICATE any part on selector module.

4.4.4 PREVENTIVE MAINTENANCE OF DIE BLOCK MODULE

Brush off any paper lint around die block entry and exit. Remove lint from punch pins and punch bar which captivates the lower ends of the punch pins.

Lubricate the following points (see Figure 4.2):

1. Die block oil reservoir, - 5 drops of oil into opening in plastic part.
2. Lower ends of punch pins where they are captivated by punch bar - 4 drops of oil total.
3. Felt pad, - oil until felt is saturated.

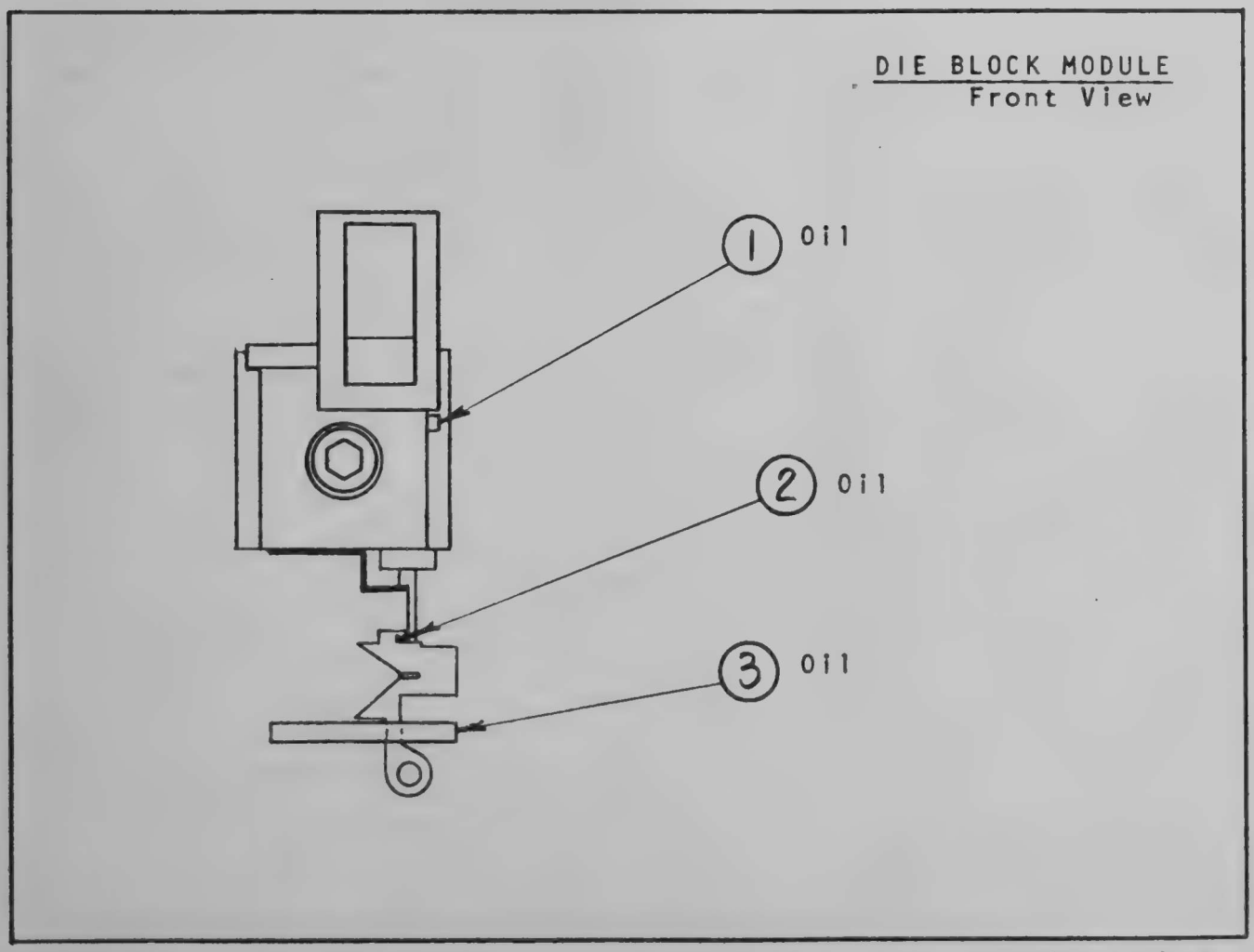


Figure 4-2

4.4.5 PREVENTIVE MAINTENANCE OF MAIN FRAME MODULE

Remove any accumulations of paper lint and/or chad from tape guide area, around feedwheel, from links below die block module, and any other area where lint has accumulated. If necessary, clean off any metallic filings or objects which might have been attracted by the magnetic pick-up.

Inspect frame assembly for worn belts, loose, or damaged parts.

Lubricate the following points (see Figure 4.3):

1. Drive motor, 2 oil holes, - 6 drops per hole.
2. Pivot link post below die block module, - 2 drops on each end of link.
3. 5 pivot points and felt washers, - 2 drops per pivot point, oil felt washers until saturation.
4. Belt tension arm bearing, - 1 drop of oil each end of bearing.
5. Belt tension roller bearing, - 1 drop of oil.

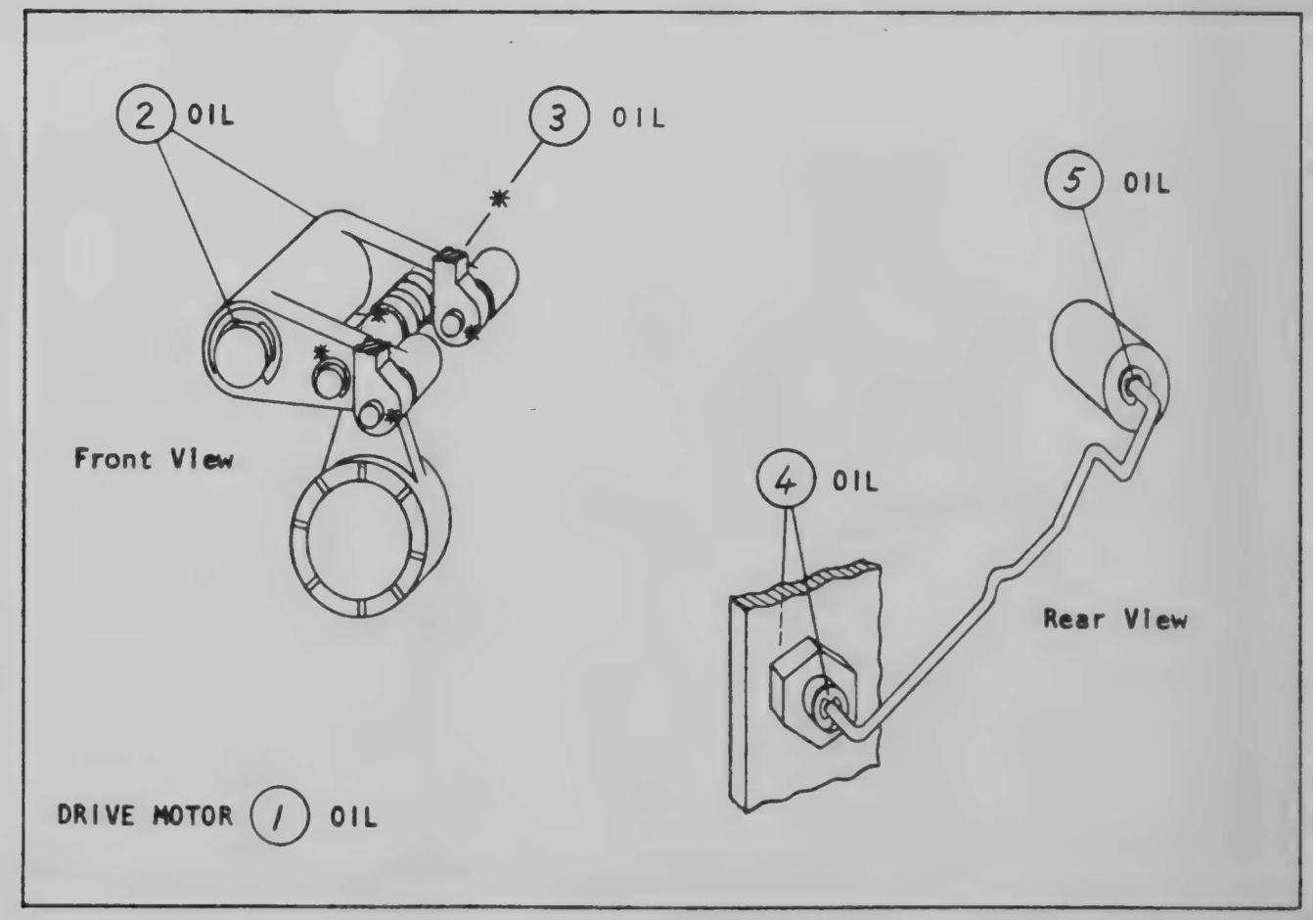


Figure 4-3

5. CORRECTIVE MAINTENANCE

5.1 GENERAL

The MODUPERF perforator mechanism is a precision instrument, carefully factory adjusted and tested. Unnecessary maintenance or attempt to improve performance must be avoided. Any adjustments found necessary should only be carried out by trained personnel equipped with the correct tools, otherwise proper operation will not be obtained.

Certain repairs and adjustments are beyond the scope of service personnel, and are, therefore, not described in this manual

CAUTION Definitely NO attempt must be made to adjust the areas/modules listed below. Return to factory for replacement.

1. Selector module, - complete unit.

2. Die block module, except tape guide (Section 5.4.6).

3. Position of mainshaft housing and pivot link post,- these parts control the location of the punch stroke (see Figure 3-1).

5.2 SEQUENCE OF ADJUSTMENTS

The sequence in which the adjustments appear should be followed when a complete readjustment of the perforator mechanism is undertaken. Since some adjustments are interrelated, making an adjustment in one area might require rechecking in other related areas. Instructions to that effect are given at the beginning of each section. In some adjustment procedures several requirements (A, B, & C) have to be met. They should be done in the given sequence.

5.3 TOOL LIST

The following are required to perform corrective maintenance.

Screw drivers; 1/32 x 1/4" blade, No. 1 and No. 2 Phillip Type.

Nut drivers; 3/16" and 11/32 hex. 5/16" hex with socket O.D. of .450" maximum.

"Allen" Head wrenches; .050", .062", 1/8", and 3/16".

6" Calipers.

Thickness Gages; .003", .005", .010", .012" and .030".

Force Gage; 280 gram (10 oz.) maximum.

5.4 DETAILED ADJUSTMENTS

5.4.1 MAGNETIC PICK-UP GAP

Interrelated affected adjustment: 5.4.2 Timing Disc Location.

Requirement A: Magnet to be positioned so that mark on magnet is nearest main panel.

Requirement B: Gap between face of magnetic pick-up and any tooth of timing disc to between .003" and .012"

To adjust:

A) Remove 3 screws, mount magnet as required.

B) Loosen 3 screws, slide pick-up in desired direction to meet requirement. Tighten screws. Re-check gap on all teeth.

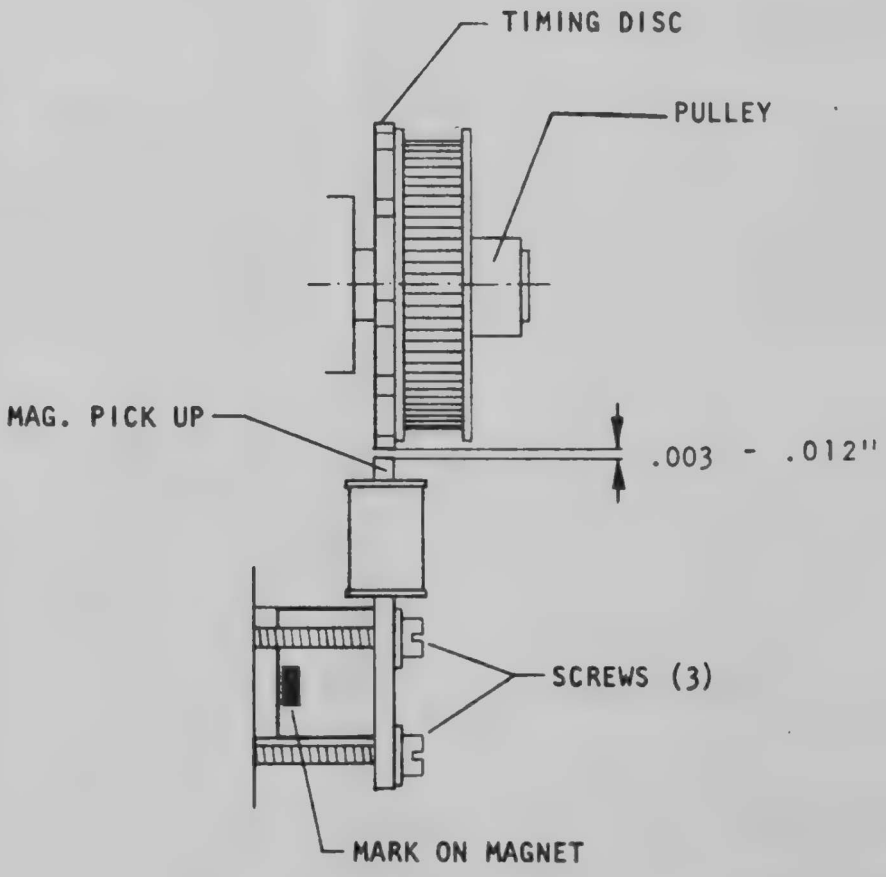


Figure 5-1

5.4.2 TIMING DISC LOCATION

Requirement A: Timing disc to line up with magnetic pick-up plate within .020", as gauged by eye.

To adjust: A: Loosen 2 set screws, move timing disc assembly in desired direction to meet requirement. Tighten screws. Re-check requirement.

Requirement B: With tooth No. 2 lined up with magnetic pick-up, slot in end of main shaft to point to 3 o'clock notch within ± 3°, as gauged by eye.

To adjust B: Loosen 2 set screws in pulley hub and rotate timing disc so No. 2 tooth lines up with magnetic pick-up. While holding timing disc in this position, rotate mainshaft (screwdriver in slot) so slot end with angled surface lines up with 3 o'clock notch in drive link. Tighten screws. Re-check requirements A and B.

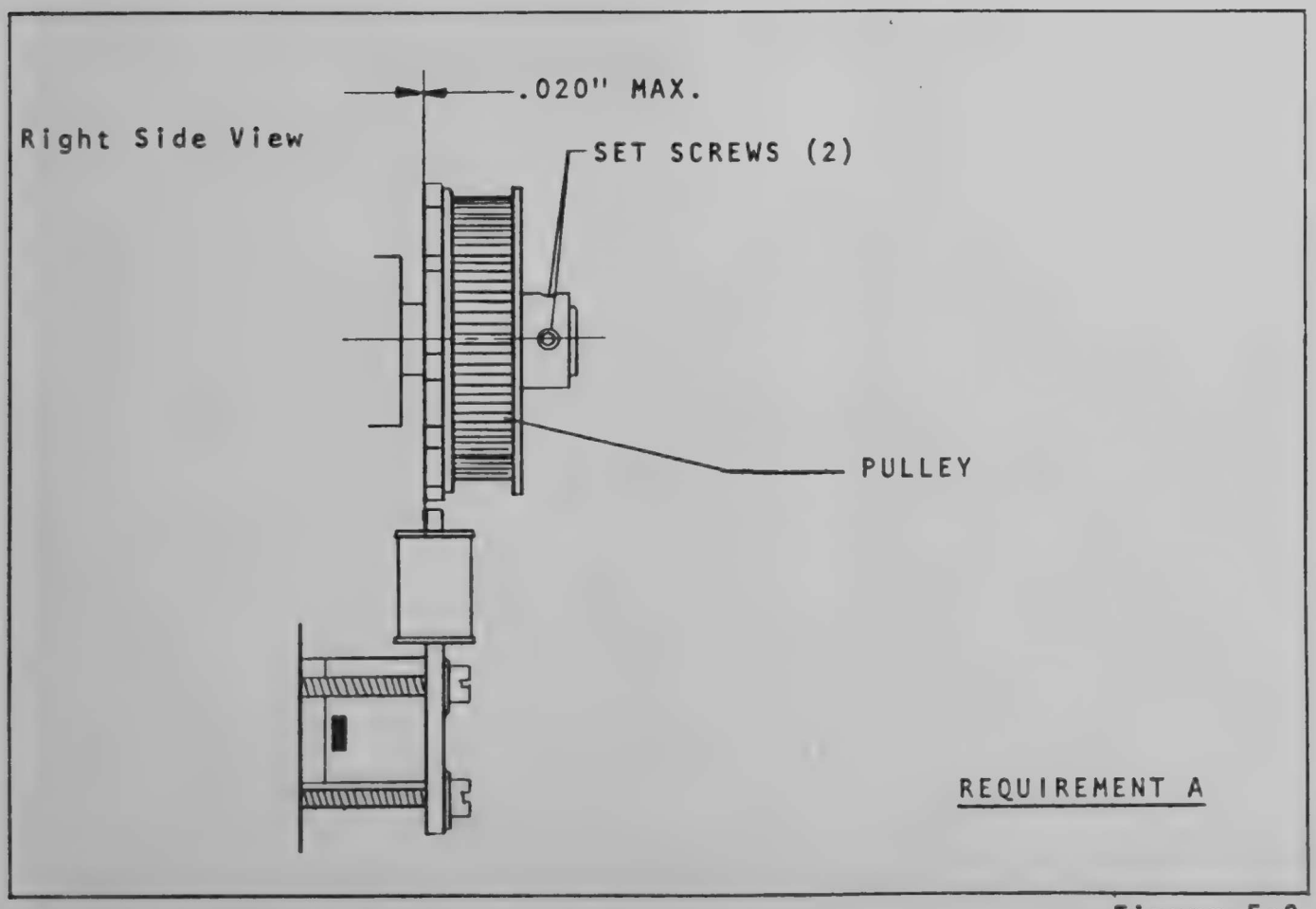


Figure 5-2

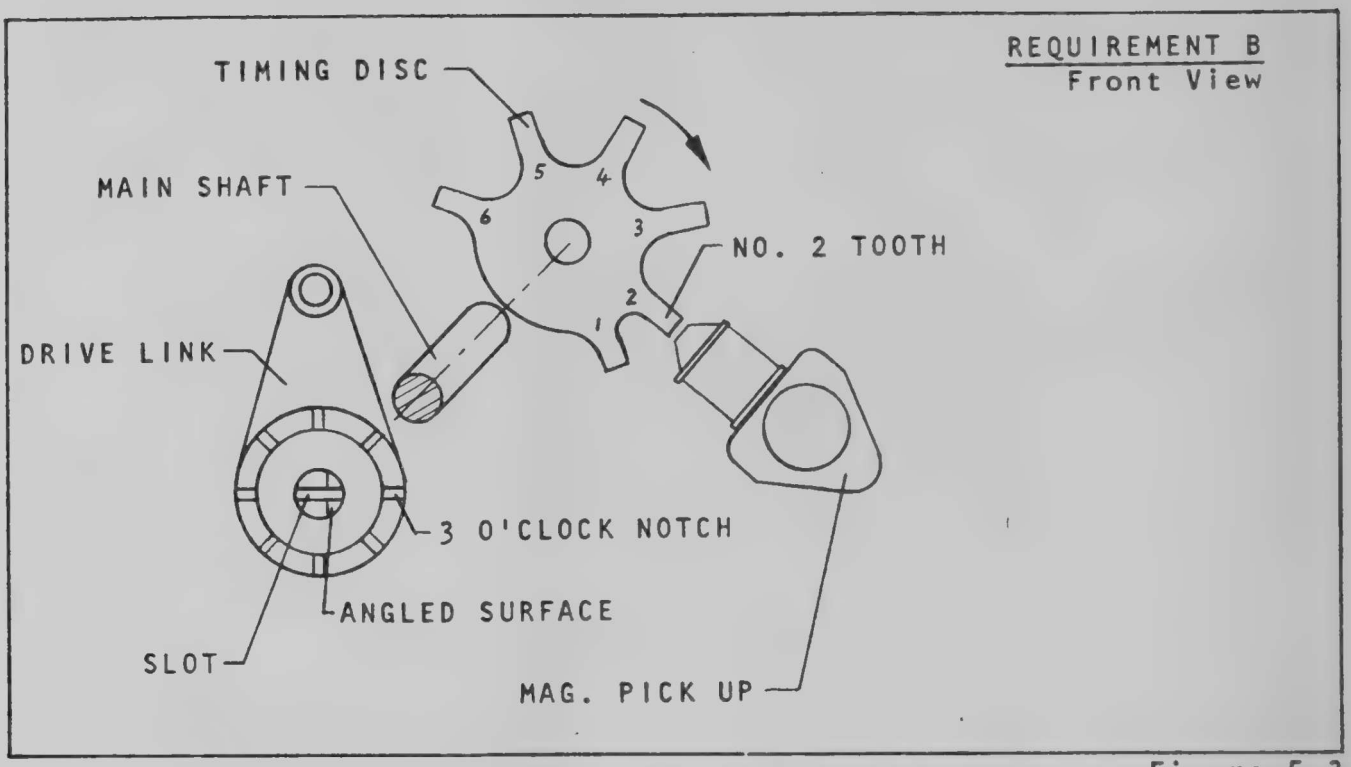

REQUIREMENT B
Front View
TIMING DISC
MAIN SHAFT
NO. 2 TOOTH
DRIVE LINK
3 O'CLOCK NOTCH
ANGLED SURFACE
SLOT
MAG. PICK UP
1
2
3
4
5
6

Figure 5-3

5.4.3 SPROCKET DRIVE ASSEMBLY

Interrelated affected adjustments: 5.4.4 Feed Sprocket Position
5.4.5 Tape Lid Gap
5.4.9 Character Spacing

The sprocket drive assembly is defined as the bearing housing, shaft, and pulley assembly which drives the feed sprocket. It is located on the immediate right of the die block module. For clarity, Figure 5-4 illustrates the sprocket drive assembly with sprocket and right top plate assembly removed.

Requirement A: Force needed to rotate pulley not to exceed 140 g (5 oz). Measure force as illustrated with BELT DISENGAGED from pulley.

Requirement B: Axial play between shaft and bearing housing not to exceed .002".

To adjust: A & B: Remove right top plate assembly (2 screws on rear of main panel) and feed sprocket (1 set screw). Loosen set screw in collar, move collar to meet requirement A and B. Tighten screw, re-check requirements.

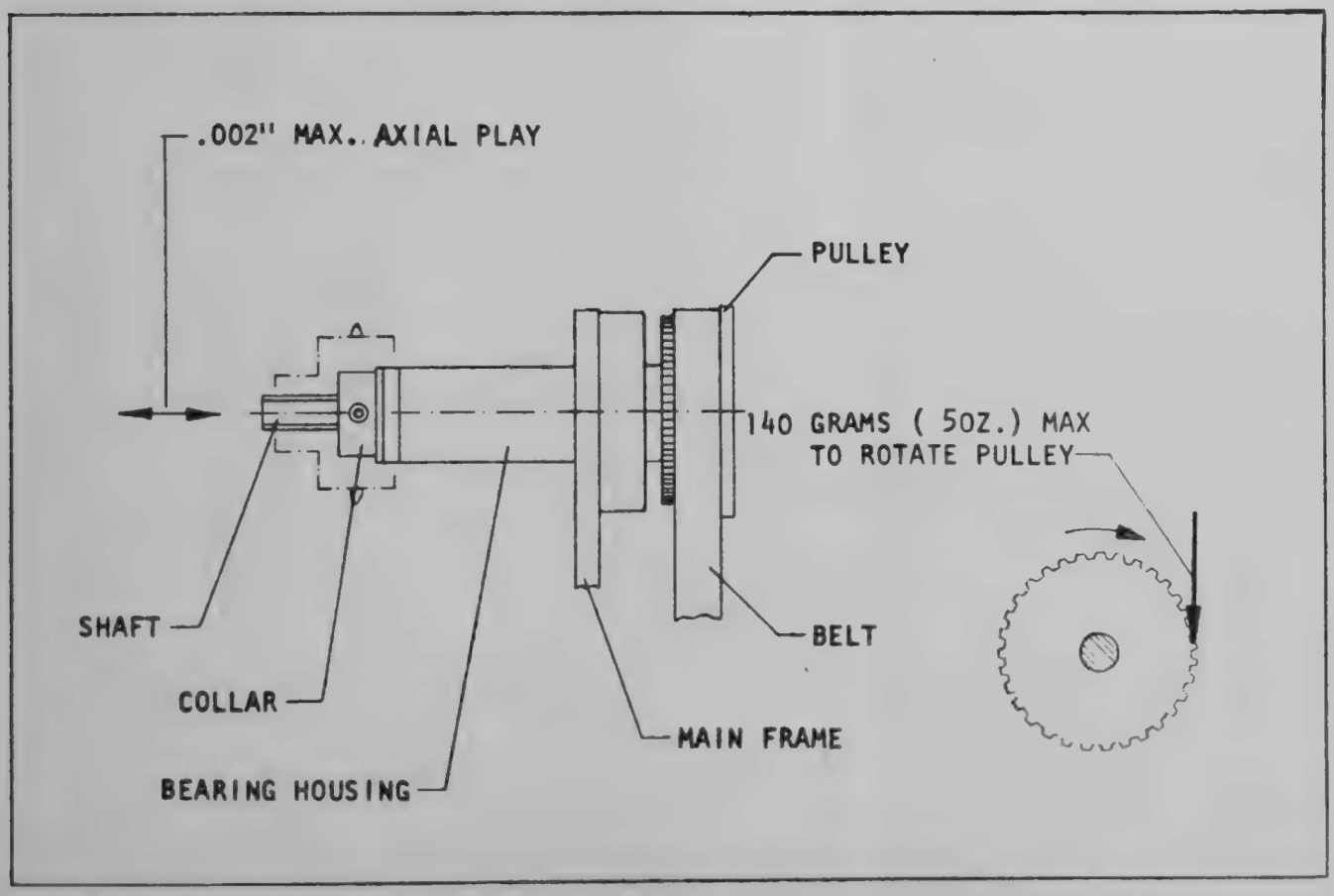


Figure 5-4

5.4.4 FEED SPROCKET POSITION

Interrelated affected adjustments: 5.4.5 Tape Lid Gap
5.4.9 Character Spacing

Requirement: Distance between front surface of main frame and center of sprocket teeth to be 1.100 ± .002".

To adjust: Remove right top plate assembly (2 screws on rear of main panel). Loosen set screw, move sprocket in desired direction, tighten set screw, re-check requirement.

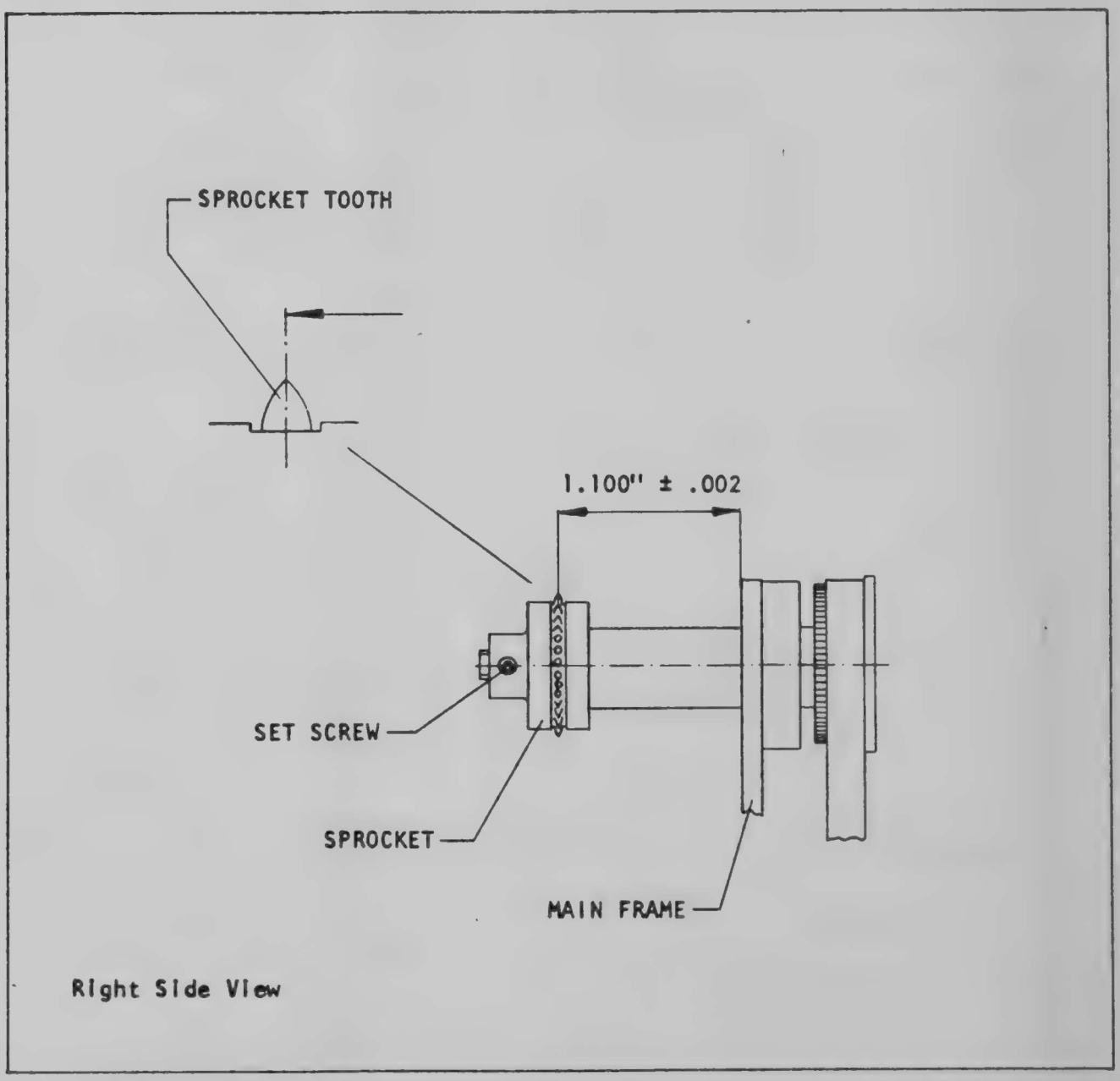


Figure 5-5

5.4.5 TAPE LID GAP

Interrelated affected adjustments: 5.4.9 Character Spacing.

Requirement: Gap between sprocket wheel and tape lid to be .005-.012". This applies to a 30° segment only, as illustrated. Outside this segment, gap to be .005" minimum. Gap between tape lid and top plate to be .010"-.030".

To adjust: Turn two adjusting screws to increase or decrease gap to meet requirement.

Note: Line up tape lid sideways so slot is centered with sprocket teeth, as gauged by eye.

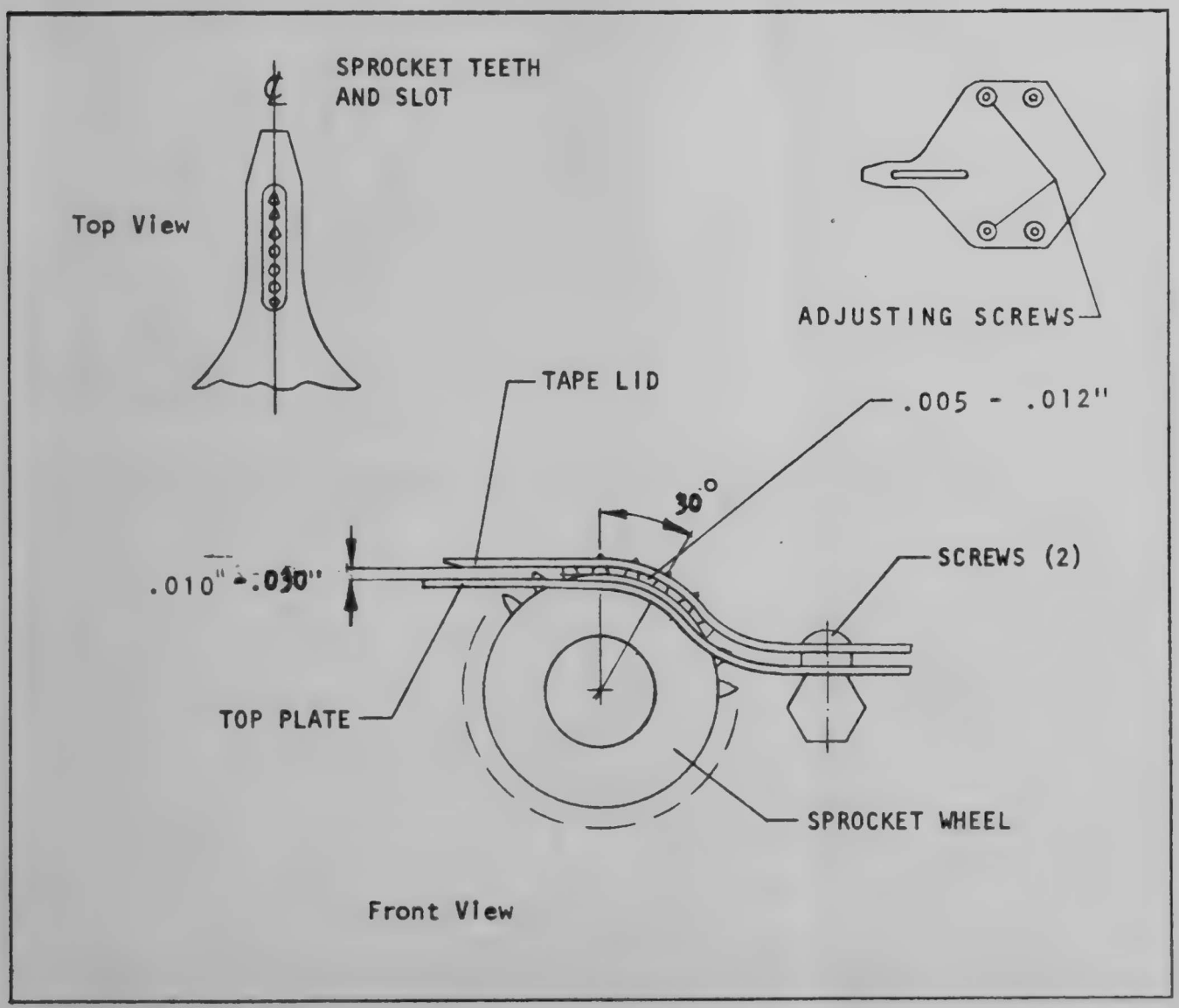


Figure 5-6

5.4.6 TAPE GUIDE

The tape guide is part of the die block module and is located left of the chad chute. A MODUPERF might be set up to punch 4 different tape configurations by changing tape guides and/or die blocks. Each tape guide has an identification number.

Requirement: Centerline of feed hole to edge of tape dimension (Dimension X, see Figure 5-7)

TAPE	GUIDE ID	X DIM.
8 Level, In Line	8	.392 ± .004"
5 Level, In Line	5	.296 ± .004"
6 Level, In Line	6,7	.392 ± .004"
6 Level, Advanced	6A	.441 ± .004"

To adjust: Loosen 2 screws which hold tape guide to die block. Insert one thickness of paper tape (.004" thick) between tape guide and frame plate. Slide guide against frame plate and tighten 2 screws. Re-check requirement. If X dimension is out of range, adjust guide with no tape or 2 thicknesses of tape.

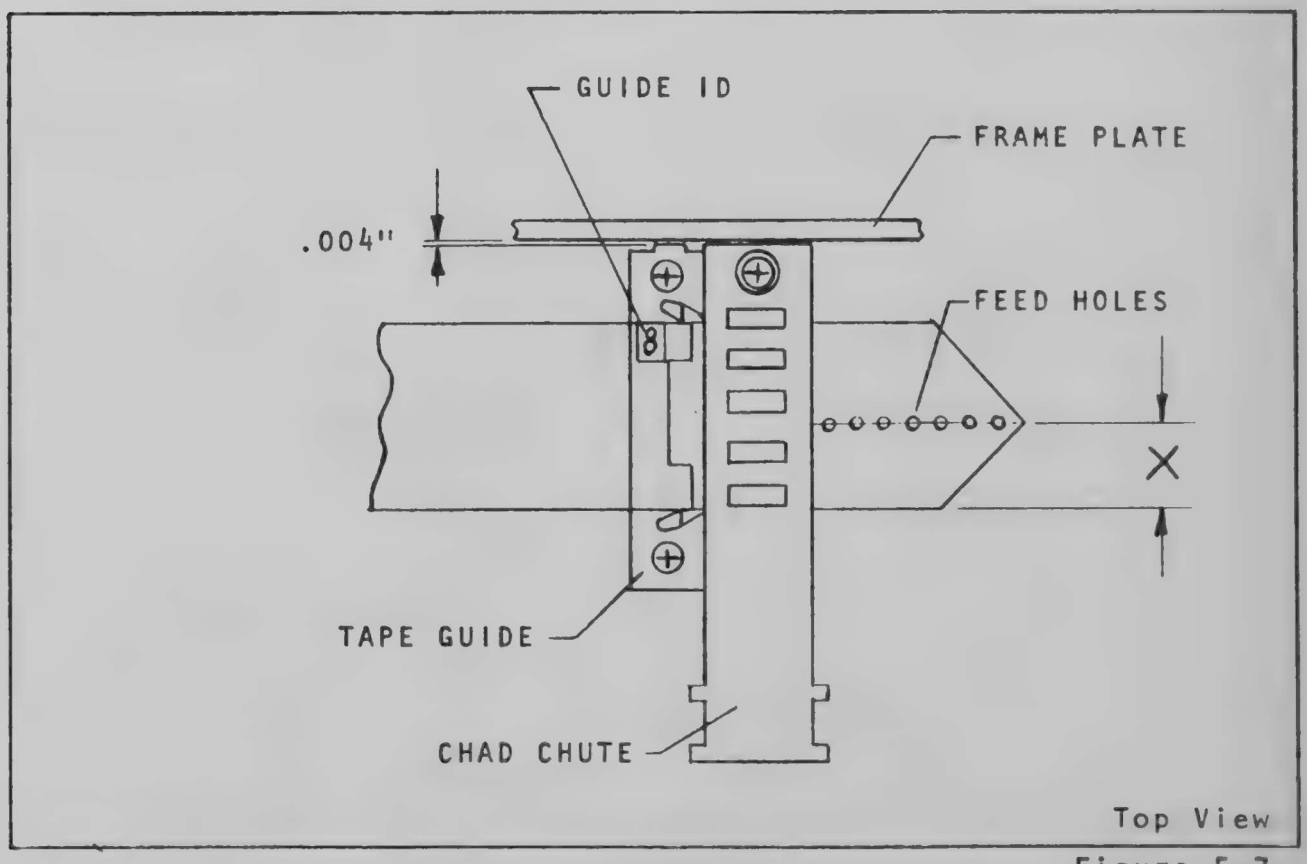


Figure 5-7

5.4.7 BELT TENSION - STEPPER MOTOR

Interrelated affected adjustments: 5.4.9 Character Spacing

Requirement: Adjust spacing between two legs of belt as illustrated. Roller force as illustrated in Figure 5-8.

To adjust: Loosen screws which hold motor mounting plate to main frame. Move motor up or down to meet belt tension requirement. Tighten screws, re-check requirement. Adjust length of spring to meet force requirement.

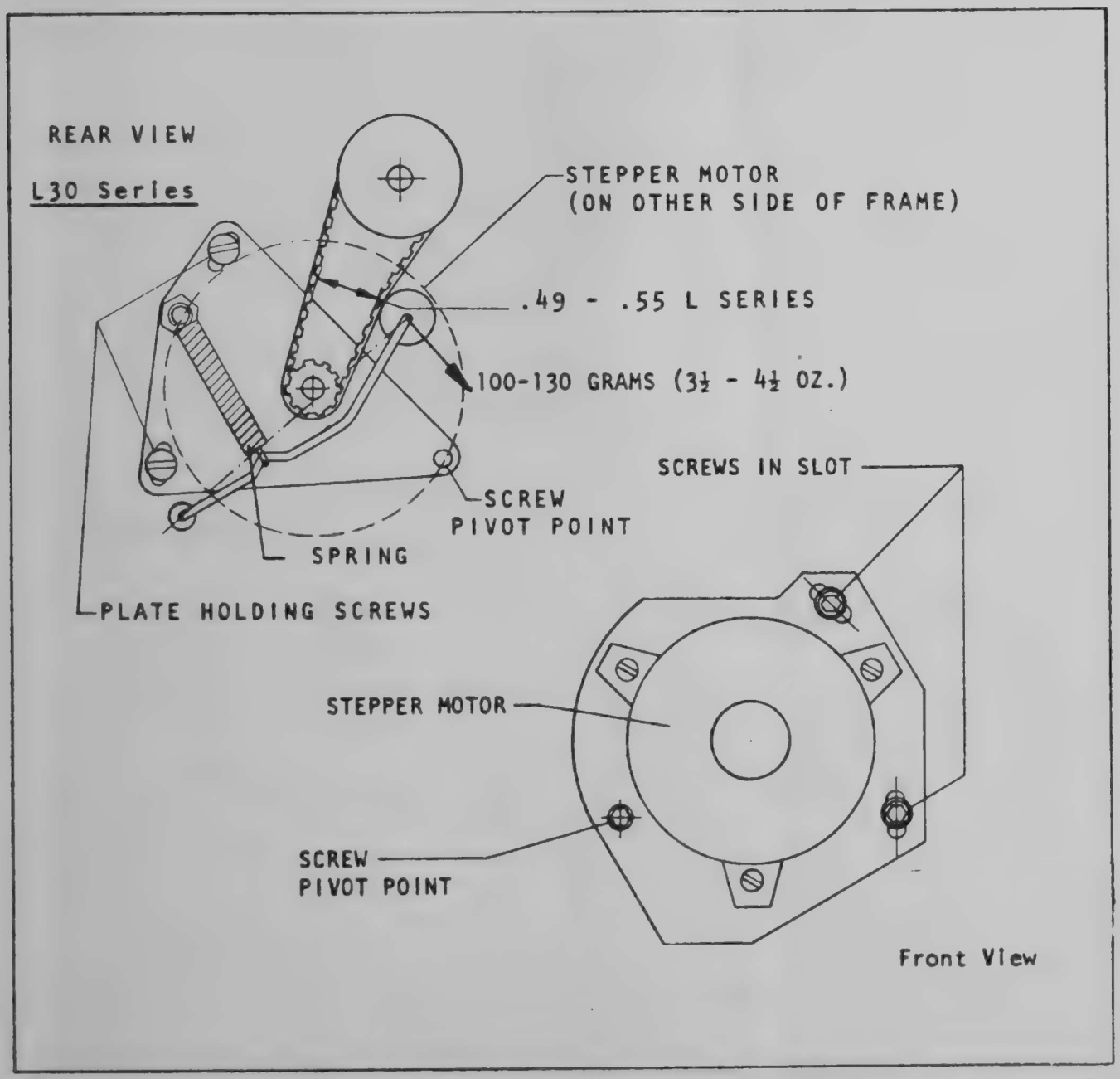


Figure 5-8

5.4.8 BELT TENSION - DRIVE MOTOR

Requirement: Timing belt to deflect specified amount with 110g (approximately 4 oz.) force applied in illustrated direction.

To adjust: Loosen motor mounting nuts and/or screws and slide drive motor left or right to meet tension requirement. DO NOT over tighten fasteners, see maximum values in figure 5-9.

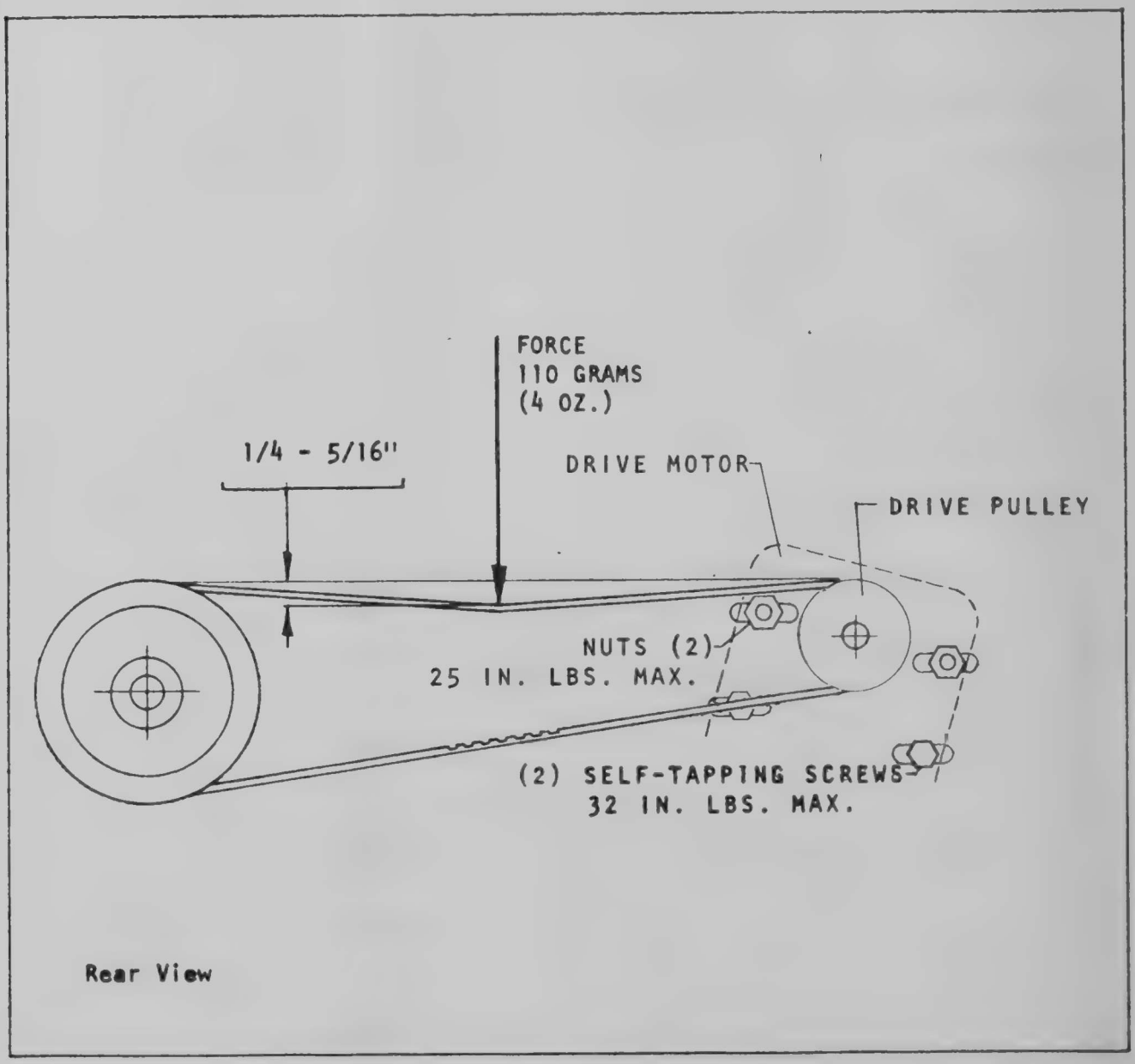


Figure 5-9

5.4.9 CHARACTER SPACING

Perforate a length of tape with a typical code arrangement. Place a feed hole over the zero line of an exact scale (scale of calipers, for example), with .1" graduations and observe the positions of the feed holes in relation to the .1" graduations.

Requirement: Cumulative spacing error in 5 inches to be a maximum of ±.025 inches.

To adjust:

Loosen motor mounting screws just enough so stepper motor can be rotated manually. Rotate clockwise for longer, counter clockwise for shorter spacing. When requirement is met, tighten screws, re-check spacing.

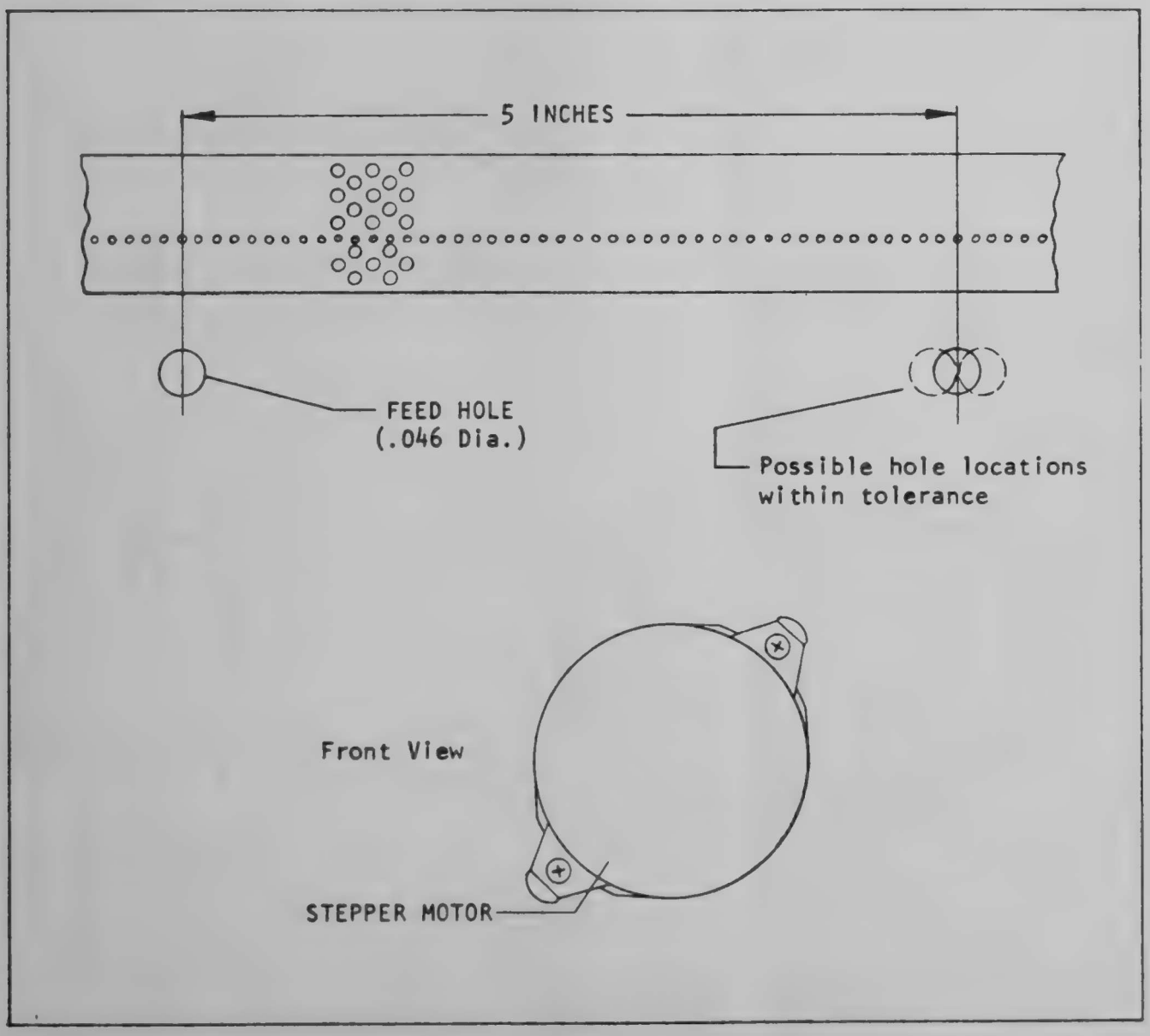


Figure 5-10

5.5 TAPE SUPPLY REQUIREMENT

In order to achieve optimum performance of the MODUPERF mechanism, 3 oz. maximum back tension on the tape is recommended. To achieve this, the tape supply configuration has to isolate the inertia of the tape roll from the perforator. Any start/stop condition will apply a back tension of 3 oz. maximum to the tape.

While there are many ways to achieve this, the concept illustrated in Figure 5-11 has been very successful.

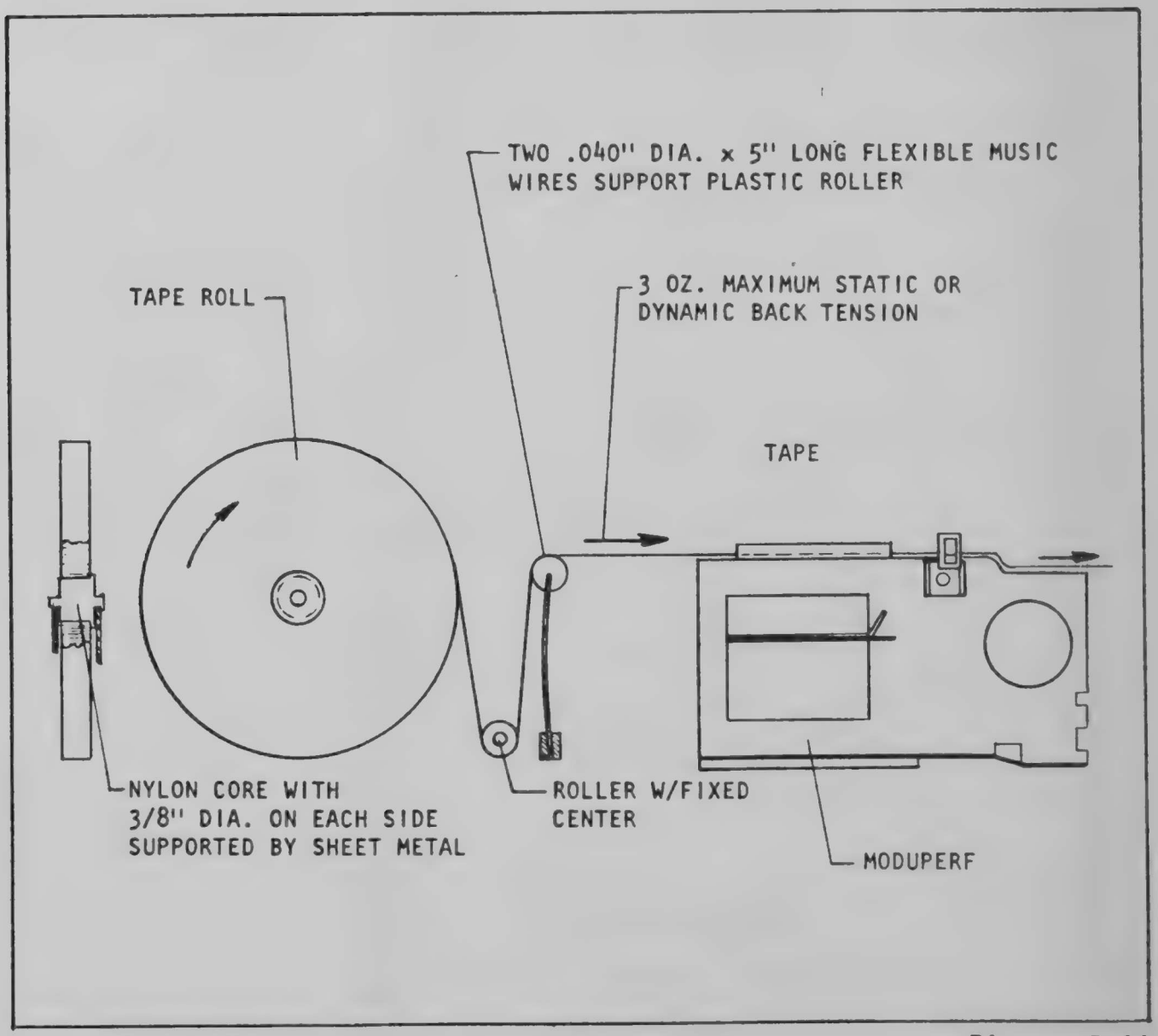


Figure 5-11